You've encountered a known limitation when using `UIDocumentInteractionController` with App Extensions on iOS. The delegate methods, such as `documentInteractionController(_:willBeginSendingToApplication:usingInteraction:)` and `documentInteractionController(_:didEndSendingToApplication:)`, are not reliably called when a user shares a file with an App Extension.

Here's a breakdown of the issue and the reasons behind it, along with references to Apple's documentation and developer discussions:

The Core Problem: A Different Process

The fundamental reason for this behavior lies in how iOS handles App Extensions. Unlike sharing with another full-fledged app, an App Extension runs as a separate, sandboxed process, and it does not launch your main (containing) application.

1. **Direct App-to-App Hand-off:** When you share a document with another application, `UIDocumentInteractionController` is designed to facilitate a direct hand-off. The system can send a message to the target app's process, and once that app is launched and receives the document, it can send a signal back to the source app. This communication allows for the delegate methods to be called.
2. **App Extension Hand-off:** In the case of an App Extension, the process is different. The user selects the extension from the `UIActivityViewController` (the share sheet). The system then launches the extension's process to handle the request. This process runs in a separate memory space and does not have a direct, real-time communication channel back to the host app (your app) in the way a traditional app-to-app hand-off does. The extension is designed to be a lightweight, focused tool to complete a specific task and then terminate.

Apple Documentation and Developer Discussions

While there isn't a single, definitive document from Apple that explicitly states "delegate methods will not be called for App Extensions," this behavior is consistent with the architectural design of App Extensions as described in Apple's documentation.

- **App Extension Programming Guide:** Apple's documentation on App Extensions emphasizes that "there is no direct communication between an app extension and its containing app; typically, the containing app isn't even running while a contained extension is running." This is a key principle. The `UIDocumentInteractionController` delegate methods rely on a communication mechanism that is not available for this type of process interaction.
- **Developer Forums:** This issue has been discussed extensively on the Apple Developer Forums and other developer communities. Many developers have observed the same behavior: the delegate methods fire for full applications (like Mail, iBooks, etc.) but not for App Extensions (like "Save to Files" or custom share extensions). Developers often confirm that the `willBeginSendingToApplication` and `didEndSendingToApplication` methods are not called, while other methods like `documentInteractionControllerWillPresentOptionsMenu` work as expected.
- **The "Application" String:** The delegate method's parameter `application: String?` is a clue. It's meant to provide the bundle identifier of the receiving *application*. An App Extension has its own bundle identifier, but the context is that

it's an extension, not a full application launch. The system's hand-off process for extensions is designed differently.

Discussion references:

1. https://developer.apple.com/forums/thread/773436
2. https://stackoverflow.com/questions/37994597/uidocumentinteractioncontrollerdelegate-methods-not-getting-called-when-shared-t
3. https://stackoverflow.com/questions/34395034/willbeginsendingtoapplication-not-called-when-open-evernote-or-any-other-extenti
4.